MATH 128a: Homework #11

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Due by: **Wednesday, November 19, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 4.3

Solve problems 2ab, 4ab, 6ab, 8ab, 14, 19(15), 20(16), 22(18) of Section 4.3 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 2: Problems from the textbook, chapter 4.4

Solve problems 1ade, 3ade, 14 of Section 4.4 of the textbook.

In problem 14c: use our notation for the composite midpoint rule: $a = x_0$, $b = x_n$ and $\int_a^b f(x)dx = 2h\sum_{j=1}^{n/2} f(x_{2j-1}) + \frac{b-a}{6}h^2 f''(\mu)$.

Problem 3: Nonlinear equations

In order to determine a fixed point x^* of a continuously differentiable mapping ϕ with $|\phi'(x)| \neq 1$ let us define the following iterative procedures for $k = 0, 1, 2, \ldots$:

(I) $x_{k+1} := \phi(x_k)$,

(II) $x_{k+1} := \phi^{-1}(x_k)$.

Show that at least one of the two iterations is locally convergent (i.e. it converges if the starting value x_0 is chosen sufficiently close to x^*.)

Problem 4: Interpolation

(a) Write down the Newton and Lagrange forms of the quadratic interpolant $p(x)$ to a function f at three points a, b and c.

(b) Give a formula for the error $p(x) - f(x)$ if f is a smooth function with all derivatives bounded. Explain why your error formula makes sense in terms of dimensions, zeroes, and the derivatives which appear versus the degree of polynomial used.

(c) Evaluate the coefficients in the Newton polynomial for $f(x) = R - 1/x$.

(d) Use (c) to express p in the form $p(x) = q_0 + q_1 x + q_2 x^2$. (i.e. find q_0, q_1, q_2)

Problem 5: Numerical differentiation

Consider the formula

$$D_0 f(x_1) := \frac{f(x_2) - f(x_0)}{2h},$$

to approximate $f'(x_1)$, where $x_2 - x_1 = x_1 - x_0 = h > 0$. Compute a bound on the error and the order of accuracy using Taylor's theorem.

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